# Module 3: Conditional Probability — Practice

**Lesson 2 | Calculate, justify, and interpret**

## Table of contents


Preparation . . . . . 1
1. Which reference group? . . . . . 1
2. Independence and exclusion . . . . . 1
3. Two selections . . . . . 1
4. Supplier evidence . . . . . 2


## Preparation

Read Lesson 2. Use its Restaurant events and supplier model inputs, retaining their stated business settings and assumptions. Refer to the English–Arabic introductions there; no additional observations are provided here.

## 1. Which reference group?

Using the Restaurant CSV, calculate $P(A \mid B)$, $P(B \mid A)$, and $P(A \cap B)$. Show the numerator and denominator for each. Interpret each in one sentence.

## 2. Independence and exclusion

1. Compare $P(A \cap B)$ with $P(A)P(B)$. Are the events independent for uniform selection from these records?
2. Explain why this is not an independence hypothesis test for a larger population.
3. Explain why Very Good and Excellent ratings are mutually exclusive but are not independent when both have positive probability.

## 3. Two selections

Find the probability that two records are both Very Good when sampling without replacement. Compare it with independent selection with replacement. State the changed second-selection denominator and numerator.

## 4. Supplier evidence

Use supplier shares 0.65 and 0.35 and bad-part rates 0.02 and 0.05 from the lesson.

1. Calculate each supplier-and-bad joint probability and the total probability of a bad part.
2. Calculate both supplier probabilities conditional on a bad part and check that they sum to 1.
3. Calculate the probability a part comes from supplier 2 **given that it is good**.
4. Explain why 0.05 is not the answer to the question “Which supplier produced this bad part?”
5. State what raw inspection information is unavailable.

Submit a table of results with reference groups and assumptions. Return to the lesson.